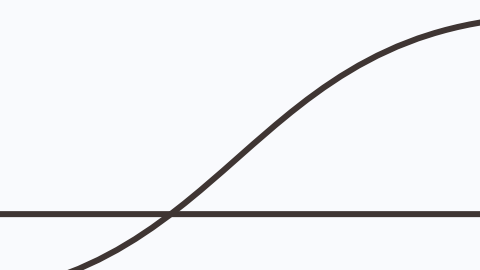


FinGuard AI: Smart Financial Policy Advisor

Under the guidance of Mr. Rithvik



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Introduction

Financial organizations manage large volumes of policy documents that are often complex and difficult to understand. Employees face challenges in quickly locating relevant information, as these policies are lengthy and written in formal language. This leads to time-consuming manual searches and inefficiency. Therefore, there is a need for a system that enables fast and accurate retrieval of policy information.

Objectives

- To develop an intelligent system for retrieving financial policy information
 - To reduce the time required to search large and complex documents
 - To provide accurate and relevant answers to user queries
 - To implement a Retrieval-Augmented Generation (RAG) based approach
 - To build a user-friendly interface for easy interaction
-

System Architecture

User Query : User enters a question through the interface

Embedding Model:
Converts the query into vector format

Retrieve & Display: Fetches relevant data and shows the final answer

Flask Backend:
Processes the request and handles communication

Vector Database:
Searches for similar information using embeddings



Workflow and Technologies used

Workflow

1. Load dataset
2. Clean text data
3. Convert into documents
4. Split into chunks
5. Generate embeddings
6. Store in vector database
7. User query → similarity search → answer

Technologies used

- Python
 - LangChain
 - Chroma Vector Database
 - Flask (Backend)
 - HTML, CSS (Frontend)
 - HuggingFace Embeddings
-

Dataset and Implemetation

Dataset:

- Financial services business dataset
- Contains company descriptions and services
- Used as knowledge base for the system

Implementation:

- Built embeddings using sentence-transformers
 - Stored data in Chroma DB
 - Implemented Flask backend for query handling
 - Created user interface for interaction
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Output

- User enters query
 - System retrieves relevant policy information
 - Displays answer quickly
 - Reduces search time significantly
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FinGuard AI

Smart Financial Policy Advisor

Ask about financial services...

Ask

FinGuard AI

Smart Financial Policy Advisor

what is finance?

finance business including purchasing and retaining eligible loans and securities guaranteeing the payment of principal and interest on securities that represent interests in or obligations secured by pools of eligible loans servicing eligible loans and issuing longterm standby purchase commitments for designated eligible loans it also engages in the purchasing and guaranteeing of securities issue

Ask about financial services...

Ask

Advantages and Limitations

Advantages:

- Fast information retrieval
- Easy to use interface
- Reduces manual effort
- Scalable system

Limitations:

- Output depends on dataset quality
 - No advanced reasoning
 - May return incomplete answers
-

Future Enhancements

- **Add LLM for intelligent responses:** Integrate a Large Language Model to generate more accurate, context-aware, and human-like responses.
 - **Improve dataset cleaning:** Enhance data preprocessing techniques to ensure higher quality, consistency, and reliability of retrieved information.
 - **Deploy as web application:** Host the system on a cloud platform to enable real-time access and scalability for multiple users.
 - **Add user authentication system:** Implement secure login and user management features to control access and protect sensitive data.
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Conclusion

In conclusion, the FinGuard AI system successfully demonstrates how modern AI techniques can be used to simplify access to complex financial policy information. By integrating data preprocessing, embedding models, and a vector database, the system enables efficient retrieval of relevant information based on user queries. The use of a Flask-based web interface ensures smooth interaction between the user and the backend system.

Overall, FinGuard AI provides a scalable and practical solution for improving information accessibility in financial organizations, with strong potential for further enhancement using advanced language models and deployment techniques.

Thank you
